Notes: Bagwell and Staiger (AER, 1999)
An Economic Theory of GATT

Contents

1 Big picture

- **Research question.** What economic problem is GATT designed to solve, and why do its two core legal principles - reciprocity and nondiscrimination - make sense as rules for trade negotiations?
- **Main answer.** GATT is interpreted as a rules-based institution that helps governments undo the inefficiency created by unilateral tariff setting. The inefficiency comes from a terms-of-trade externality: each government can shift part of the cost of protection onto foreign exporters by changing world prices.
- **Core theoretical move.** The paper allows governments to have general political-economy objectives, not just national-income maximization. Even with political motives, the only inefficiency that trade agreements need to solve is the terms-of-trade-driven cost-shifting externality.
- **Role of reciprocity.** Reciprocity means reciprocal tariff changes that produce balanced changes in import volumes. In the model, such changes leave world prices unchanged. This neutralizes terms-of-trade manipulation and makes governments behave as if they did not value world-price movements.
- **Role of nondiscrimination.** In a multicountry world, discriminatory tariffs create an additional local-price externality because an importing country cares which foreign supplier provides imports when tariff rates differ across suppliers. MFN/nondiscrimination removes this extra externality, leaving only the world-price externality that reciprocity can handle.
- **Role of preferential agreements.** Free-trade areas and many customs unions violate nondiscrimination. They can therefore revive local-price externalities and weaken the efficiency properties of the multilateral system.
- **Key takeaway.** Reciprocity and nondiscrimination are not merely mercantilist or legalistic norms. They are simple negotiation rules that jointly guide governments toward politically optimal, nondiscriminatory, and efficient tariff outcomes.

2 Incremental contribution of the paper

- **A unified economic theory of GATT.** Earlier work studied optimum tariffs, tariff wars, or reciprocal liberalization, but this paper offers a unified framework to interpret GATT's foundational rules.
- **Separating political motives from agreement motives.** The paper allows government preferences to include distributional concerns. It then shows that political motives affect the desired local price, but do not by themselves create the need for a trade agreement. The agreement is needed because unilateral tariff choices manipulate world prices.
- **Political optimum concept.** The paper introduces the "politically optimal" tariff as the tariff a government would choose if it cared about local-price effects but did not value the terms-of-trade consequences of its tariff. This concept becomes the benchmark efficient target of GATT rules.
- **Reciprocity as a world-price-neutral rule.** The paper formalizes the idea that reciprocal tariff changes leave world prices unchanged. This makes reciprocity an economic device for neutralizing cost-shifting incentives.

- **Renegotiation-proof interpretation.** Under GATT Article XXVIII-style renegotiation rules, efficient agreements are implementable under reciprocity if and only if tariffs are politically optimal. This gives a dynamic/renegotiation-based explanation for why reciprocity selects a specific efficient point.
- **Nondiscrimination as a complement to reciprocity.** In the multicountry model, reciprocity alone cannot generally deliver efficiency when tariffs are discriminatory. MFN ensures that all relevant externalities travel through world prices, restoring the effectiveness of reciprocity.
- **Regionalism critique.** The paper provides a theoretical reason why preferential agreements can undermine multilateral cooperation: they are discriminatory and can therefore reintroduce local-price externalities.
- **Rules-based versus power-based institutions.** The paper argues that GATT's rules can reduce the role of bargaining power, making participation more attractive to weaker countries.

3 Conceptual framework

3.1 Economic environment

The baseline model is a two-country, two-good, perfectly competitive general equilibrium trade model. The home country naturally imports good x and exports good y; the foreign country naturally imports y and exports x.

Let

$$p = \frac{p_x}{p_y}, \qquad p^* = \frac{p_x^*}{p_y^*}, \qquad p^w = \frac{p_x^*}{p_y},$$

where p and p^* are local relative prices and p^w is the world untaxed relative price. Domestic and foreign ad valorem import tariffs are represented by

$$\tau = 1 + t, \qquad \tau^* = 1 + t^*.$$

The local prices satisfy

$$p = \tau p^w, \qquad p^* = \frac{p^w}{\tau^*}.$$

Thus, a domestic tariff increase raises the domestic local price and lowers the world price, improving the domestic terms of trade but worsening the foreign terms of trade.

3.2 Production, consumption, and market clearing

Production is determined by local prices:

$$Q_i = Q_i(p), \qquad Q_i^* = Q_i^*(p^*) \quad \text{for } i \in \{x, y\}.$$

Consumption depends on local prices and tariff revenue:

$$C_i = C_i(p, p^w), \qquad C_i^* = C_i^*(p^*, p^w).$$

Domestic imports of x are

$$M_x(p, p^w) = C_x(p, p^w) - Q_x(p),$$

and domestic exports of y are

$$E_y(p, p^w) = Q_y(p) - C_y(p, p^w).$$

The equilibrium world price is pinned down by market clearing, for example by the y-market condition

$$E_y(p(\tau, p^w), p^w) = M_y^*(p^*(\tau^*, p^w), p^w).$$

3.3 Government preferences

Rather than assuming national-income maximization, the paper represents government objectives generally as

$$W(p(\tau, p^w), p^w), \qquad W^*(p^*(\tau^*, p^w), p^w).$$

The only imposed structure is that, holding the local price fixed, each government prefers an improvement in its own terms of trade:

$$W_{p^w} < 0, \qquad W^*_{p^w} > 0.$$

This formulation nests national-income maximization and political-economy models in which governments value sectoral or distributional outcomes. The model therefore distinguishes two motives:

- **local-price motive:** how the government values the domestic allocation and distribution implied by p;
- **world-price motive:** how the government values shifting costs through terms-of-trade movements.

3.4 Nash tariffs

When each government chooses its tariff unilaterally, the home best response satisfies

$$W_p \frac{dp}{d\tau} + W_{p^w} \frac{dp^w}{d\tau} = 0.$$

Dividing by $dp/d\tau$ gives

$$W_p + \lambda W_{p^w} = 0,$$

where

$$\lambda = \frac{dp^w/d\tau}{dp/d\tau} < 0.$$

The foreign best response is analogous:

$$W^*_{p^*} + \lambda^* W^*_{p^w} = 0.$$

The key insight is that unilateral tariff choices combine the local-price motive with the world-price motive. Because each government benefits from shifting some costs onto its trading partner, Nash tariffs are too protectionist relative to the efficient level.

3.5 Politically optimal tariffs

The politically optimal tariffs are defined by stripping out the terms-of-trade motive:

$$W_p = 0, \qquad W^*_{p^*} = 0.$$

These are the tariffs each government would choose if it cared about domestic political and economic effects but not about world-price cost shifting. If governments maximize national income, politically optimal tariffs are reciprocal free trade. With political-economy motives, politically optimal tariffs need not be zero, but they are efficient because they remove only the inefficient international externality.

4 Main propositions and intuition

4.1 Proposition 1: Nash tariffs are inefficient

Nash tariffs are inefficient because each government ignores the harm that its tariff-induced world-price movement imposes on its trading partner. The domestic tariff lowers the world price of the imported good, shifting part of the protection cost to foreign exporters. The foreign country behaves symmetrically. The result is excessive protection on both sides.

4.2 Proposition 2: A reciprocal trade agreement entails reciprocal liberalization

Starting from Nash tariffs, a mutually beneficial reciprocal trade agreement must lower both countries' tariffs. This follows because each country is already choosing a best response to the other country's tariff. A unilateral increase in one country's tariff cannot make both countries better off; the only way to produce mutual gains is to reduce protection together.

4.3 Proposition 3: Politically optimal tariffs are efficient

Politically optimal tariffs satisfy $W_p = 0$ and $W^*_{p^*} = 0$. These conditions mean each government chooses its preferred local price, holding aside the terms-of-trade motive. Once both governments ignore cost shifting, the resulting tariff pair lies on the efficiency frontier.

4.4 Proposition 4: Reciprocity can generate monotonic welfare gains from Nash

A reciprocal tariff reduction from Nash that leaves world prices unchanged raises both governments' welfare until at least one government reaches its preferred local price at the initial world price. If countries are symmetric, this path leads exactly to the politically optimal outcome.

The intuition is simple. At Nash, each government would like more trade if it could avoid a terms-of-trade deterioration. Reciprocity makes this possible by balancing tariff concessions so that world prices remain unchanged.

4.5 Proposition 5: Under reciprocity, efficient implementable agreements are politically optimal

When GATT-style renegotiation is allowed and renegotiation must preserve reciprocity, the only efficient agreement that survives renegotiation is the politically optimal tariff pair. If an efficient agreement is not politically optimal, at least one government wants to move along the same world-price locus to a preferred local price. Reciprocity permits such a renegotiation. Only at the political optimum does neither government want to renegotiate.

4.6 Proposition 6: In a multicountry world, politically optimal tariffs are efficient if and only if they satisfy MFN

With more than two countries, discriminatory tariffs create local-price externalities. If the home country charges different tariffs to different exporters, then it cares not only about total imports but also about which foreign country supplies them. The local prices of foreign exporters affect the composition of trade and therefore domestic tariff revenue. MFN removes this distortion by treating all foreign suppliers equally.

4.7 Proposition 7: Reciprocity plus MFN implements efficient multilateral agreements

In the multicountry model, an efficient multilateral trade agreement can be implemented under reciprocity if and only if tariffs both satisfy MFN and are set at their politically optimal levels. Reciprocity neutralizes world-price externalities; MFN ensures that no additional local-price externalities remain.

4.8 Proposition 8: Free-trade areas are incompatible with efficient multilateral implementation

Free-trade areas are discriminatory because members eliminate internal barriers while retaining tariffs against nonmembers. This violates MFN, revives local-price externalities, and prevents reciprocity from generally delivering multilateral efficiency.

4.9 Proposition 9: Customs unions are compatible only under narrow conditions

A customs union eliminates internal barriers and adopts a common external tariff. Such a union can be treated like a single country only if members are natural integration partners, meaning the internal allocation of revenue and prices is efficient for the member countries. Otherwise, customs unions also threaten multilateral efficiency.

5 Detailed interpretation of reciprocity

5.1 Reciprocity as balanced concessions

The paper defines reciprocity as mutual tariff changes that generate equal changes in import volumes across trading partners, measured at existing world prices. In the two-country model, this condition implies

$$[p_1^w - p_0^w]M_x(p(\tau_1, p_1^w), p_1^w) = 0.$$

Since trade is positive, reciprocity implies

$$p_1^w = p_0^w.$$

Thus, reciprocal concessions leave world prices unchanged.

This gives a precise interpretation of the GATT norm of balanced concessions. Negotiators are not simply following mercantilist confusion. Instead, they are using reciprocity to prevent unilateral liberalization from worsening their terms of trade.

5.2 Krugman's "GATT-think" reconsidered

Krugman summarizes trade-negotiation logic as: exports are good, imports are bad, and balanced increases in imports and exports are good. Bagwell and Staiger reinterpret these rules:

- **Exports are good** because another country's tariff cut improves the home country's terms of trade.
- **Imports are bad** because the home tariff cut worsens the home country's terms of trade relative to unilateral best response.
- **Balanced import and export expansion is good** because reciprocity neutralizes the world-price loss while expanding trade volume.

5.3 Reciprocity in renegotiation

Under GATT Article XXVIII, a country may seek to raise a previously bound tariff. If it does so, trading partners may withdraw substantially equivalent concessions. In the model, this means the renegotiation preserves the original world price. Any agreement that leaves one government wanting less trade at the prevailing world price will be renegotiated. Hence, reciprocity does not merely help tariff reductions; it also disciplines tariff increases.

6 Detailed interpretation of nondiscrimination

6.1 Why MFN is unnecessary in the two-country case

In the two-country model, there is only one foreign supplier and one world price. No discrimination across suppliers is possible. Therefore, reciprocity alone is enough to neutralize the relevant externality.

6.2 Why MFN matters in the multicountry case

With multiple foreign suppliers, the home country may impose different tariffs on different partners. If it does, the home government cares about the composition of imports, not only the total import volume. For a fixed total volume, imports from the higher-tariff partner generate more tariff revenue. Thus, foreign countries' local prices affect the home country's welfare by changing trade shares.

This creates a local-price externality. Reciprocity is designed to neutralize world-price effects, not local-price effects. Therefore, reciprocity alone is insufficient when tariffs are discriminatory.

MFN solves the problem by requiring equal treatment of all foreign suppliers. Once all suppliers face the same tariff, the importing country's welfare no longer depends on the identity of the supplier in the same distortionary way. Externalities again travel through world prices only, and reciprocity can operate effectively.

7 Preferential agreements and regionalism

7.1 Free-trade areas

A free-trade area eliminates tariffs among members but allows each member to keep its own external tariff against nonmembers. This is inherently discriminatory: inside partners receive preferential treatment, while outsiders do not. In the paper's framework, free-trade areas are fundamentally at odds with the multilateral logic of reciprocity plus nondiscrimination.

7.2 Customs unions

A customs union eliminates internal tariffs and adopts a common external tariff. This is less problematic than a free-trade area because the union can sometimes be treated as a single country. However, this works only if internal allocation is efficient. The paper calls such countries natural integration partners. This condition requires sufficiently aligned preferences and income levels.

7.3 Main implication for WTO/GATT design

Article XXIV permits preferential agreements, but the paper argues that this exception is costly from the perspective of multilateral efficiency. Preferential agreements can revive the very externalities that reciprocity and nondiscrimination are supposed to eliminate.

8 Rules-based versus power-based negotiation

The paper contrasts two institutional designs:

- **Power-based bargaining:** governments bargain directly over tariffs, and outcomes depend heavily on relative bargaining power.
- **Rules-based bargaining:** governments negotiate under rules such as reciprocity and nondiscrimination, which limit the bargaining chips that can be used.

GATT is rules-based in two senses. First, negotiations occur under agreed principles. Second, those principles reduce the role of power asymmetries. Reciprocity narrows the feasible bargaining set and pushes outcomes toward the political optimum, which is defined without reference to relative bargaining power.

This can encourage participation by weaker countries. A weak country might refuse to participate in power-based negotiations if it expects exploitation after making sunk investments to serve a large country's market. A reciprocity-based system can make participation safer by limiting the strong country's ability to use bargaining power to extract concessions.

9 Figures

9.1 Figure 1: The world- and local-price effects of a tariff change

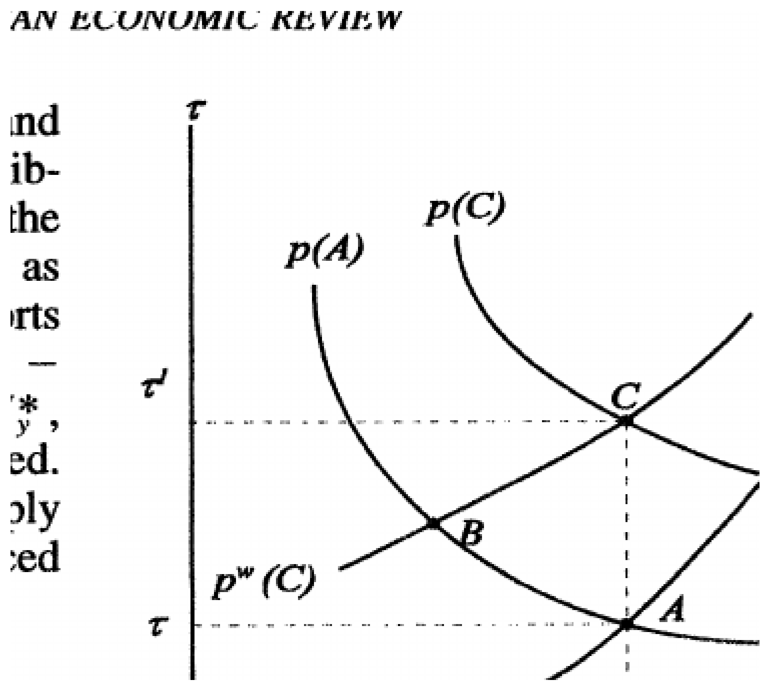


Figure 1: The world- and local-price effects of a tariff change.

Read:

- The figure decomposes a tariff-induced movement into two parts: a world-price effect and a local-price effect.
- Moving from A to B isolates the improvement in the domestic terms of trade: the world price falls while the local price is held fixed.
- Moving from B to C captures the local-price effect: the domestic local price rises, changing the domestic allocation.
- This decomposition is the key to the paper's theory. Political motives can affect how governments value the local-price movement, but the inefficient international externality comes from the world-price movement.

Economic message: A tariff does two things: it changes domestic allocation and it shifts costs internationally. Trade agreements are needed only for the second channel.

9.2 Figure 2: The purpose of a reciprocal trade agreement

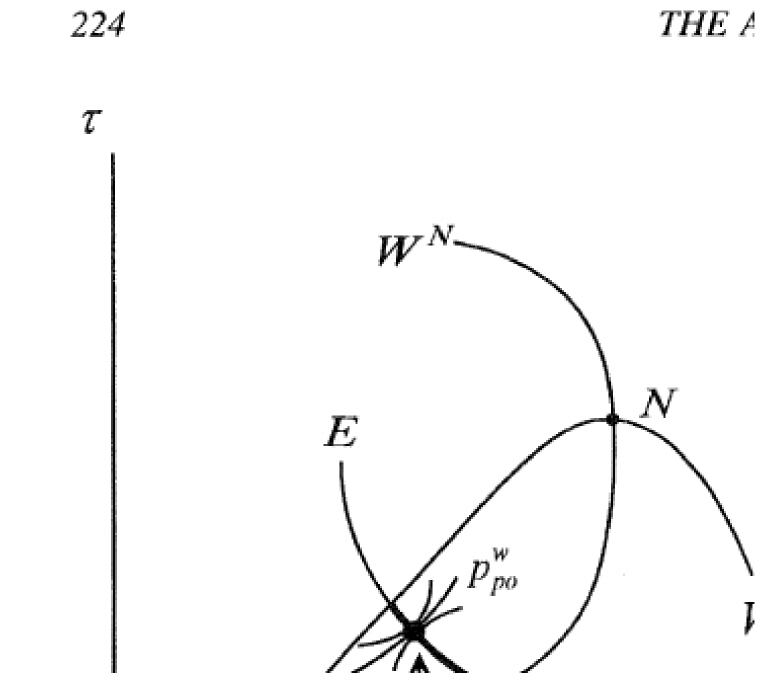


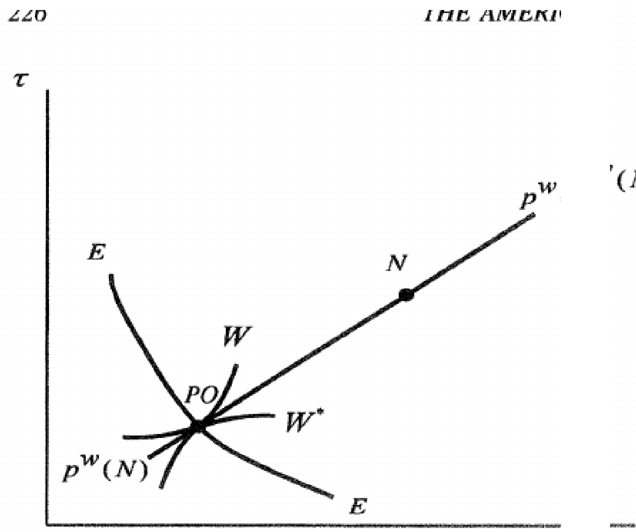
Figure 2: The purpose of a reciprocal trade agreement.

Read:

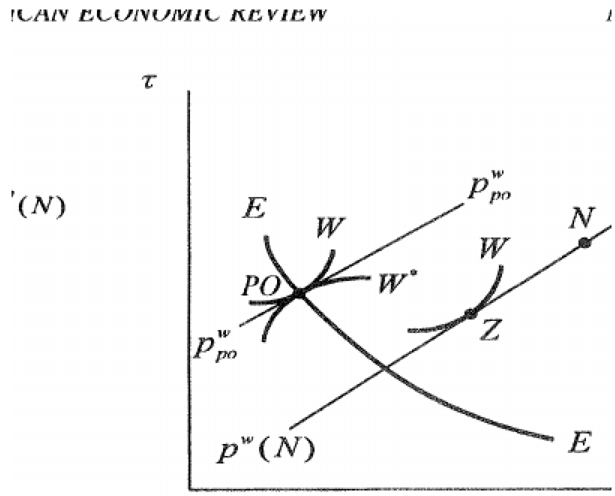
- N is the Nash tariff outcome. It lies off the efficiency frontier because unilateral tariffs are distorted by terms-of-trade manipulation.
- E–E is the efficiency locus. Along this locus, no government can be made better off without making another worse off.
- PO is the politically optimal point. At this point, governments achieve preferred local prices without using tariffs for world-price cost shifting.
- The bold part of the efficiency locus is the contract curve, where both governments can gain relative to Nash.

Economic message: The purpose of a reciprocal trade agreement is to move countries from inefficient Nash protection toward the efficiency frontier, especially toward politically optimal tariffs.

9.3 Figures 3A and 3B: Liberalization according to reciprocity



(a) Figure 3A: Symmetric case.



(b) Figure 3B: Asymmetric case.

Figures 3A and 3B: Liberalization according to reciprocity.

Read:

- Figure 3A shows the symmetric case. Reciprocal liberalization along the Nash iso-world-price locus reaches the politically optimal point.
- Figure 3B shows the asymmetric case. Reciprocal liberalization initially benefits both countries, but the path may stop at point Z before reaching the efficiency frontier, because one country reaches its preferred local price first.
- In both cases, reciprocity works by keeping the world price fixed while expanding trade.

Economic message: Reciprocity makes liberalization politically attractive because it creates more trade without imposing a terms-of-trade loss on either government.

9.4 Figure 4: Renegotiation under reciprocity

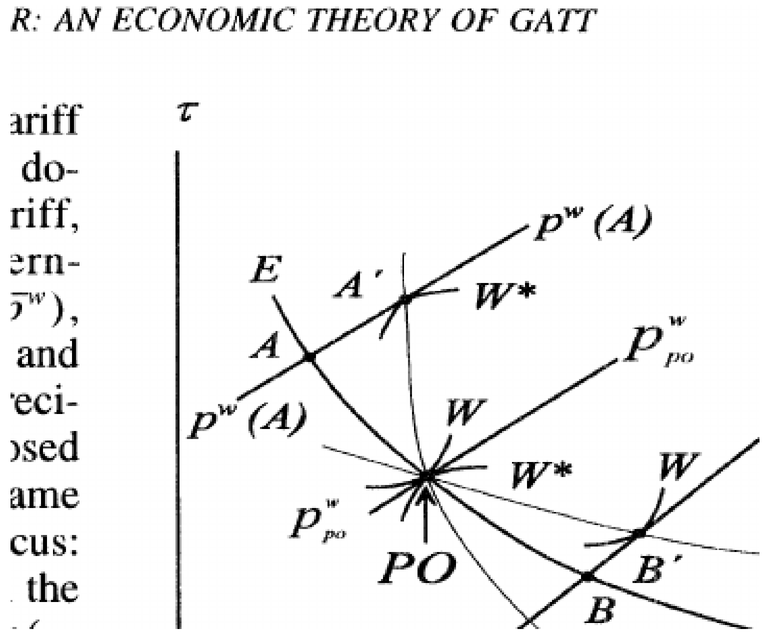


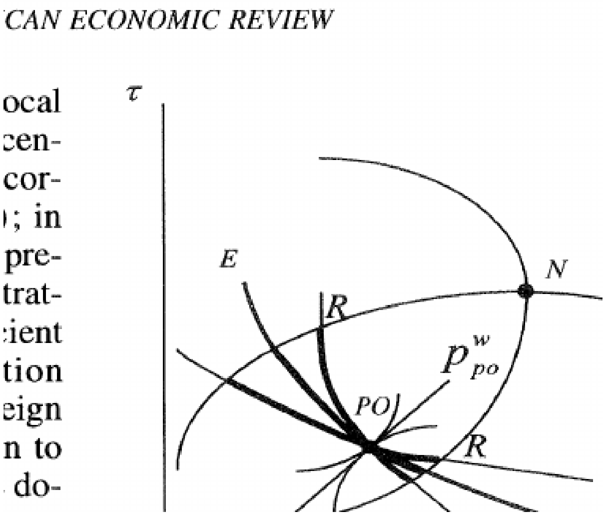
Figure 4: Renegotiation under reciprocity.

Read:

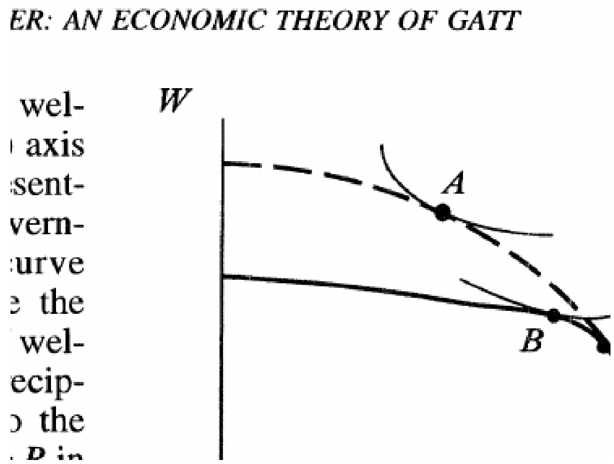
- The figure shows why efficient agreements other than PO may not be renegotiation-proof.
- At point A, the foreign government wants to move along the same iso-world-price locus to A', where it achieves its preferred local price.
- At point B, the domestic government wants to move to B'.
- Only at PO does neither government wish to move along the world-price-preserving locus.

Economic message: GATT-style reciprocity in renegotiation selects politically optimal tariffs as the unique efficient renegotiation-proof outcome.

9.5 Figures 5A and 5B: Implementable bargaining outcomes under reciprocity



(a) Figure 5A: Tariffs implementable under reciprocity.



(b) Figure 5B: Welfare outcomes implementable under reciprocity.

Figures 5A and 5B: Implementable outcomes under reciprocity.

Read:

- Figure 5A shows the set of tariff pairs implementable under reciprocity. The implementable set is restricted relative to the full efficiency frontier.
- Figure 5B translates this into welfare space. The dashed curve is the unconstrained efficiency frontier, while the bold curve is the reciprocity-constrained set.
- The reciprocity-constrained bargaining solution B is closer to the political optimum than the unconstrained power-based solution A.

Economic message: Reciprocity changes the bargaining environment. It reduces the scope for power-based extraction and pushes outcomes toward the political optimum.

10 Short summary

- The paper develops a general equilibrium theory of GATT.
- Governments may have political-economy preferences, but the international inefficiency from unilateral tariffs comes from terms-of-trade cost shifting.
- Politically optimal tariffs are the tariffs governments would choose if they did not value world-price manipulation.
- Nash tariffs are inefficient because each government shifts some protection costs onto trading partners.
- Reciprocity keeps world prices fixed during reciprocal tariff changes, neutralizing terms-of-trade manipulation.
- In two-country settings, reciprocity can guide governments toward politically optimal tariffs.

- In multicountry settings, nondiscrimination/MFN is needed because discriminatory tariffs create local-price externalities.
- Reciprocity plus MFN can implement efficient multilateral trade agreements.
- Free-trade areas generally undermine this system; customs unions are compatible only under narrow natural-integration conditions.
- GATT's rules-based approach can reduce the importance of bargaining power and encourage participation by weaker countries.

11 Conclusion

11.1 Core conclusion

Bagwell and Staiger argue that the economic purpose of GATT is to remove the terms-of-trade externality created by unilateral tariff setting. The principles of reciprocity and nondiscrimination are not arbitrary legal norms. They are mechanisms that help governments implement efficient trade agreements despite political motives and bargaining constraints.

11.2 Economic intuition

A tariff is attractive unilaterally because it raises the domestic local price and improves the domestic terms of trade. But the terms-of-trade gain is a cost imposed on foreign exporters and foreign governments. When every country behaves this way, protection is excessive. Reciprocity solves the problem by ensuring that mutual tariff changes do not alter world prices. Nondiscrimination ensures that, in a multicountry world, no additional local-price externalities arise from discriminatory treatment of suppliers.

11.3 Incremental contribution revisited

The paper's contribution is not simply to say that trade agreements reduce tariffs. It explains why the legal architecture of GATT has economic content. Reciprocity turns tariff bargains into world-price-neutral exchanges. MFN makes reciprocity effective by preventing discriminatory local-price spillovers. Political optimality provides the target point: the tariff structure governments would choose without the inefficient cost-shifting motive.

11.4 Implications

For trade theory, the paper establishes the terms-of-trade approach as a central rationale for trade agreements. For trade policy, it warns that preferential agreements may weaken the multilateral system. For institutional design, it suggests that rules can discipline bargaining power and make cooperation more attractive to weaker countries.

11.5 Limitations

The model abstracts from enforcement constraints, domestic commitment motives, security and geopolitical motives for regional agreements, and institutional details of actual WTO dispute settlement. The empirical section is suggestive rather than a direct structural test. Still, the theoretical framework became foundational because it gives a coherent economic interpretation of GATT's core principles.

11.6 Takeaway

The enduring takeaway is that GATT's architecture can be understood as an efficiency device: reciprocity neutralizes terms-of-trade manipulation, nondiscrimination removes discriminatory local-price externalities, and together they guide governments toward efficient multilateral trade policy.